

Pension Fund Challenges and AI/ML Solutions: A Governance Framework in Progress

The 2025 Autumn Governance Series: Pension Fund Governance Focus



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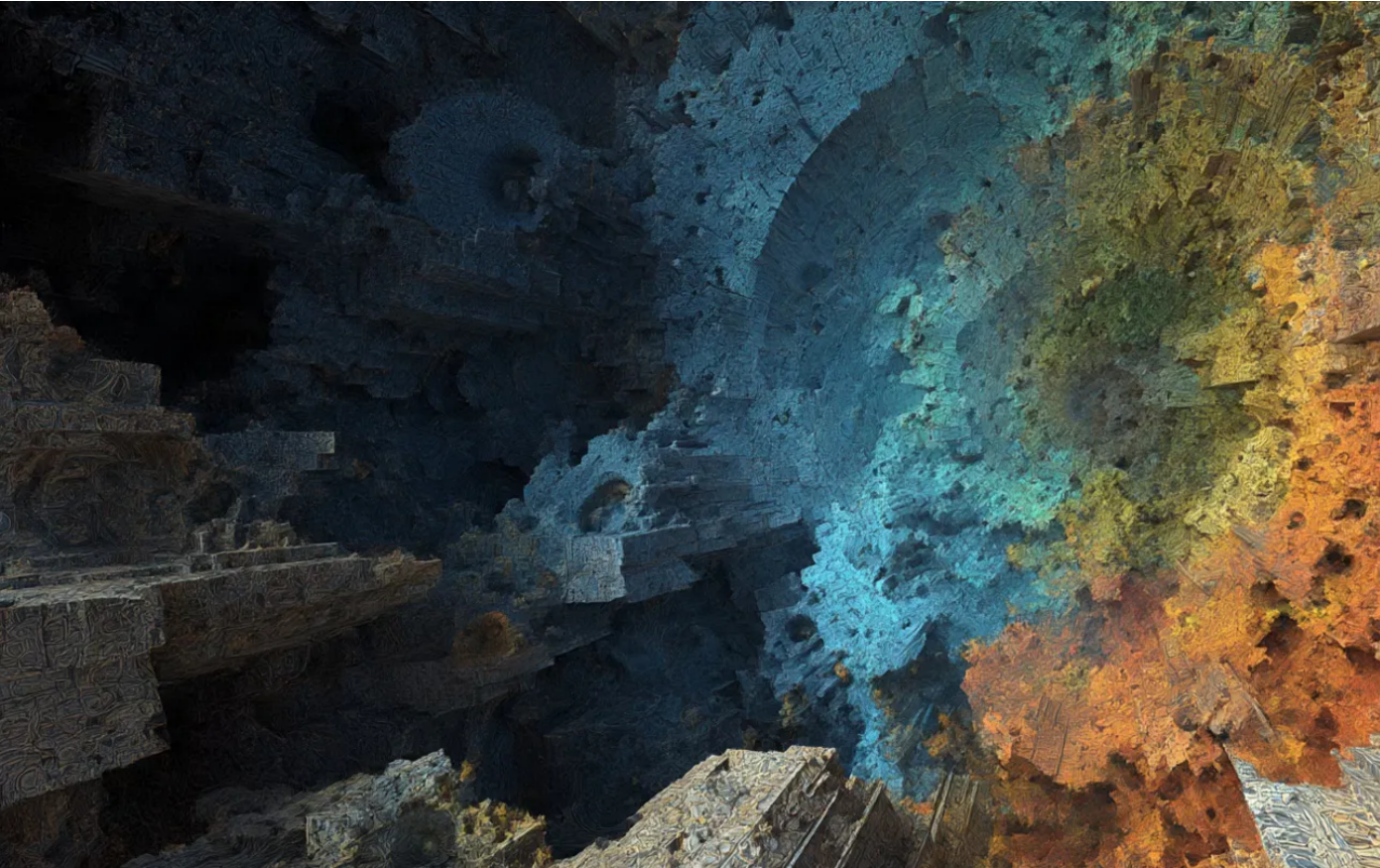
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Pension systems are living archives of society's promises—how we trade decades of work into dignity. Yet across jurisdictions, those promises are under strain. Demographic collapse, climate volatility, and widening wealth gaps have exposed how fragile the traditional actuarial model has become.

As a governance practitioner, I approach this not as an AI evangelist but as someone concerned with institutional trust. The question isn't whether algorithms will shape pension oversight—they already do in adjacent spaces—but how boards can adapt governance so that technology strengthens, rather than erodes, fiduciary accountability.

Demographic and Actuarial Imbalances

The math is shifting. Worker-to-retiree ratios in OECD countries have fallen from roughly **16:1 in the 1950s to below 2:1 today**, projected to reach **1:1 by 2050** (OECD *Pensions at a Glance*, 2023). Women retire with **25–30 % of pension income**, reflecting career interruptions averaging seven years and lower lifetime wages (European Commission *Pension Adequacy Report* 2021). Mortality differences between high- and low-income groups can exceed a decade, undermining pooled assumptions of longevity.

Researchers are testing machine-learning models that integrate income, health, and geography to build **stratified mortality tables** instead of national averages. Continuous cohort updates could flag dependency-ratio shifts earlier. For governance, the issue is not coding technique but oversight: trustees validate such tools, demand explainability, and ensure policy consequences—like contribution rate changes—remain transparent to members.

Climate Risk and Long-Term Sustainability

Climate change has moved from externality to solvency variable. Defined benefit schemes in moderate warming scenarios may see **funding horizons extend by 3–9 years** and solvency ratios drop **up to 40 %** (IFoA *Planetary Solvency Framework*, 2023). Climate-related mortality and morbidity patterns further complicate actuarial projections.

AI can enrich scenario testing by merging **IPCC climate pathways** with economic and asset data to model 1.5 °C, 2 °C, or 3 °C worlds. Some funders, like **APG in the Netherlands**, are experimenting with machine-learning climate simulators that link physical risk to portfolio stress tests. The governance challenge is methodological: how to weigh probabilistic model outputs when fiduciary duty still demands prudence over prediction.

Equity and Inclusion Gaps

Equity gaps are measurable, not abstract. Women perform **75 % of unpaid care work** and hold **37 % smaller private pension pots**, generating a **\$3 trillion global shortfall** by 2030 (World Economic Forum, 2020). In the US, Black households hold **43 %**, and Hispanic households **23 %**, of white household retirement wealth (Federal Reserve, 2022).

Emerging “fairness-aware” actuarial research re-weights data to correct structural bias, testing policies such as caregiver credits or contribution caps. Singapore’s **CPF Board** has piloted analytics to identify under-savings cohorts and adjust outreach. Governance boards could borrow these insights to treat gender and racial fairness as quantifiable risk factors—integrating equity audits into their regular oversight cycle.

Private-Market Opacity and Valuation Risk

Private equity valuations remain opaque, often relying on discretionary models reviewed quarterly at best. When performance data sit behind proprietary algorithms, trustees govern assumptions, not reality.

Machine-learning **automated valuation models** are being tested by institutional investors to benchmark reported net-asset values against comparables. **CalPERS**, for example, used NLP tools to review private-contracts for fee discrepancies—useful, but it revealed the limits of algorithmic interpretation. For boards, the lesson is procedural: autom insights require documented human review, audit trails, and clear escape thresholds.

Governance and Information Asymmetry

Even strong boards face temporal and informational gaps. Liabilities stretch over decades while board terms and reporting cycles remain short. Surveys show **95 % of directors expect AI to affect their organizations**, yet few **30 % regularly discuss it** (NACD, 2024).

Explainable-AI dashboards could help visualize funding ratios, climate exposure, and demographic shifts, allowing trustees to interrogate assumptions rather than delegate them. But governance must stay ahead of automation—developing policies for model documentation, independent validation, and ethical review before such tools become routine.

Toward an AI-Ready Governance Framework

Structural Oversight

Boards should begin by **formally assigning responsibility** for AI oversight, either through a dedicated risk sub-committee or an expanded technical mandate. Recruiting one or two trustees with data, actuarial, or cybersecurity backgrounds can anchor the conversation in practical competence. Cross-functional steering groups (CIO, CRO, Chief Actuary, and tech leads) ensure implementation remains aligned with fiduciary objectives, not vendor roadmaps.

Policy and Standards

Before any algorithm is deployed, require a concise **AI Impact Assessment** documenting purpose, data sources, expected benefits, and fairness implications. Maintain a **model inventory** listing ownership, validation and known limitations. For decisions affecting members or valuations, demand **explainability outputs**—feature-importance scores, error margins, and human sign-off at key thresholds.

Measuring and Disclosing Trust

Governance increasingly depends on visibility. Boards can track metrics such as model accuracy, fairness audit results, climate exposure, and transparency rankings. A public-facing dashboard—similar to the **NYC Comptroller's pension portal**—could disclose asset allocations, climate metrics, and gender or racial gap trends. Regular reporting of model validations and trustee training builds confidence among regulators and beneficiaries alike.

Independent Validation and Continuous Learning

Independent third-party reviews of major AI models should complement financial audits, verifying data integrity, reproducibility, and bias control. Trustee education remains essential: annual “AI and Governance” workshops can help directors understand what explainability means, when to question outputs, and how to integrate ethical oversight into routine decision-making.

Implementation in Practice

Reform need not be radical.

- In **Year 1**, boards can assess readiness, establish policy, and pilot one low-risk AI use case (for instance, a chatbot for member queries).
- **Years 2–3** can focus on scaling—introducing climate or demographic analytics, commissioning an independent audit, and launching a transparency dashboard.
- By **Year 4**, AI assurance could be embedded in the annual governance cycle, with findings published alongside financial statements.

Counterarguments and Practical Limitations

AI adoption in pensions is not a simple story of progress. **Smaller funds** face high implementation costs or lack data quality to justify complex systems. Some risk creating “**governance theater**”—committees and dashboards that signal oversight but lack real accountability. Others may drift toward **quantitative over-reliance**, sidelining the judgment that fiduciary duty requires.

The **digital divide** among trustees is real: some are deeply conversant with AI ethics; others are still mastering basic digital workflows. Training budgets and time constraints make parity difficult. And while model validation frameworks sound reassuring, many rely on the same experts who built the system, creating a structural conflict of interest that regulators have yet to resolve.

These are not reasons to avoid innovation, but reminders that responsible AI adoption begins with humility about institutional capacity.

Voices from the Field

A trustee perspective—paraphrased from interviews and commentaries in the *Canadian Investment Review* and *Benefits Canada*—suggested that “AI is only as good as the governance culture that surrounds them. If trustees cannot interpret the output, the technology becomes decoration.”

A viewpoint echoed in publications from the **UK Pensions Policy Institute** and **ShareAction** emphasized that “transparency isn’t just data disclosure—it’s ensuring members understand how decisions affect them. That’s where technology can widen or narrow trust.”

Similarly, the **European Insurance and Occupational Pensions Authority (EIOPA)**, in its 2023 consultation on AI and algorithmic governance, stressed that regulators are more concerned with “quality of documentation than speed of adoption. If boards can demonstrate comprehension, oversight becomes credible.”

These voices remind us that technology governance ultimately lives or dies on interpretation and dialogue—not just compliance checklists.

Balancing Innovation and Prudence

AI and machine learning may never eliminate actuarial uncertainty, but can sharpen foresight and expose biases long hidden in aggregate data. The governance task is to turn these tools into allies of accountability—ensuring every model is interpretable, auditable, and aligned with fiduciary duty.

Ultimately, trust is still the pension system's most valuable asset. The algorithms entering its architecture should be measured not only by accuracy but by how well they reinforce that trust.

This series explores ideas from “[Shaping the Decade: Governance, Sustainability, and AI 2026–2036](#),” a guide for boards at the crossroads of governance, technology, and stakeholder capitalism. [Available Here](#).

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